4 **(a)** Fig. 4.1 shows apparatus used to observe the motion of smoke particles (Brownian motion).

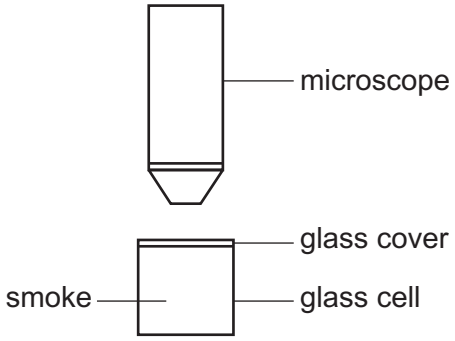


Fig. 4.1

The glass cell has light shining on it from the side.

The smoke particles are seen as bright specks of light when looking through the microscope.

(i) Draw the path of **one** of the bright specks of light.

[2]

(ii) Explain, in terms of forces and the motion of air molecules, the cause of the motion of the smoke particles.

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..... [4]

(b) The temperature of the air in a sealed glass container is increased.

(i) Explain, in terms of molecules, why the internal energy of the air increases.

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..... [1]

(ii) Explain, in terms of molecules, why the pressure of the air also increases.

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..... [2]

[Total: 9]

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